

Session 1: How can I do math?

A Slower Introduction to R

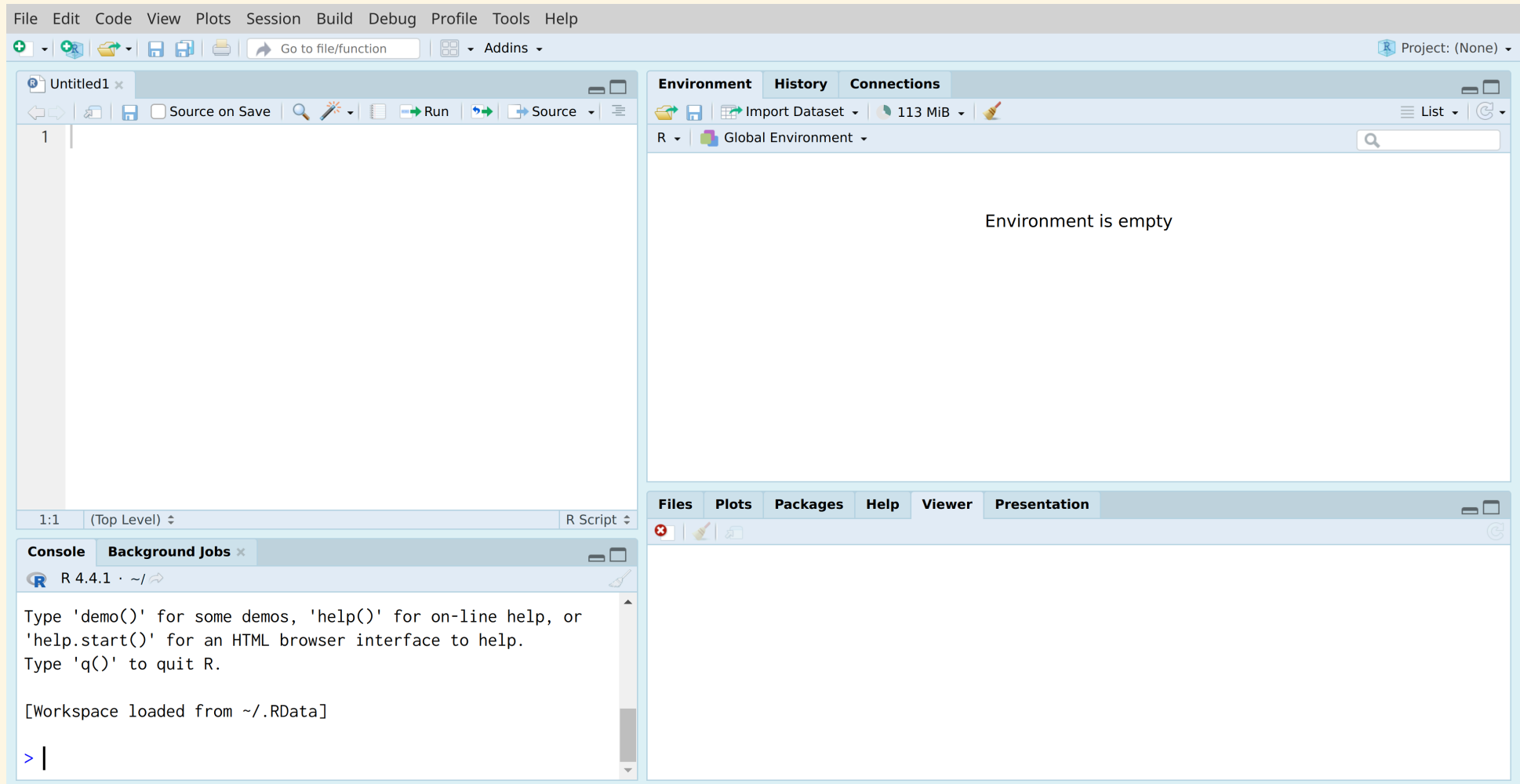
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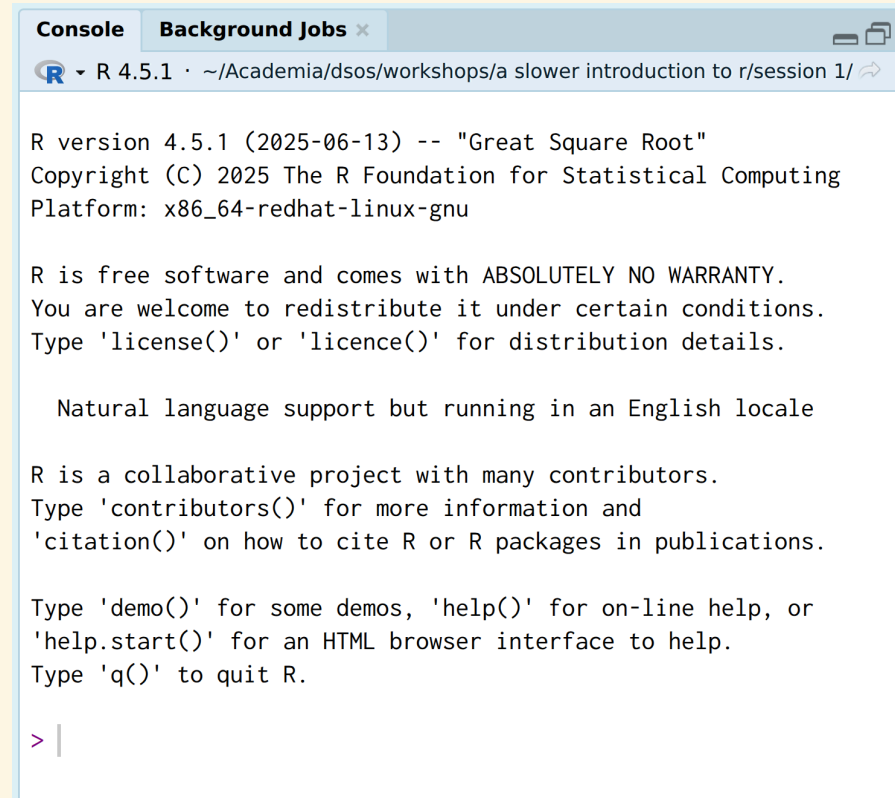
Friday, October 2, 2026

RStudio's panes

RStudio's panes



The console



The screenshot shows the R console window with the following text:

```
R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-redhat-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

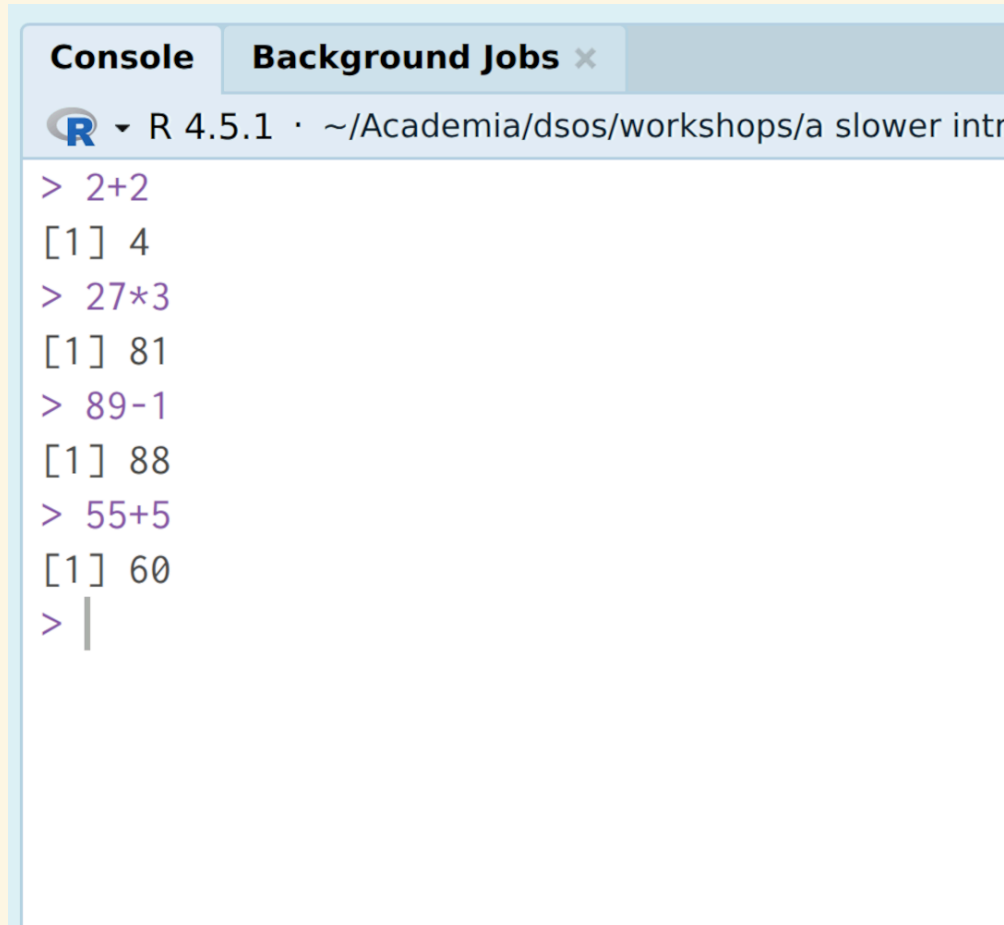
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

The greater-than sign (`>`) is called a **prompt** (“it’s your turn”), and is a space for you to write messages to your computer.

The console is like a message thread



The screenshot shows an R console window with two tabs: 'Console' and 'Background Jobs'. The console displays a series of commands and their outputs, mimicking a message thread. The commands are entered in purple, and the outputs are in black. The path shown is `~/Academia/dsos/workshops/a slower intr`.

```
R 4.5.1 · ~/Academia/dsos/workshops/a slower intr
> 2+2
[1] 4
> 27*3
[1] 81
> 89-1
[1] 88
> 55+5
[1] 60
> |
```



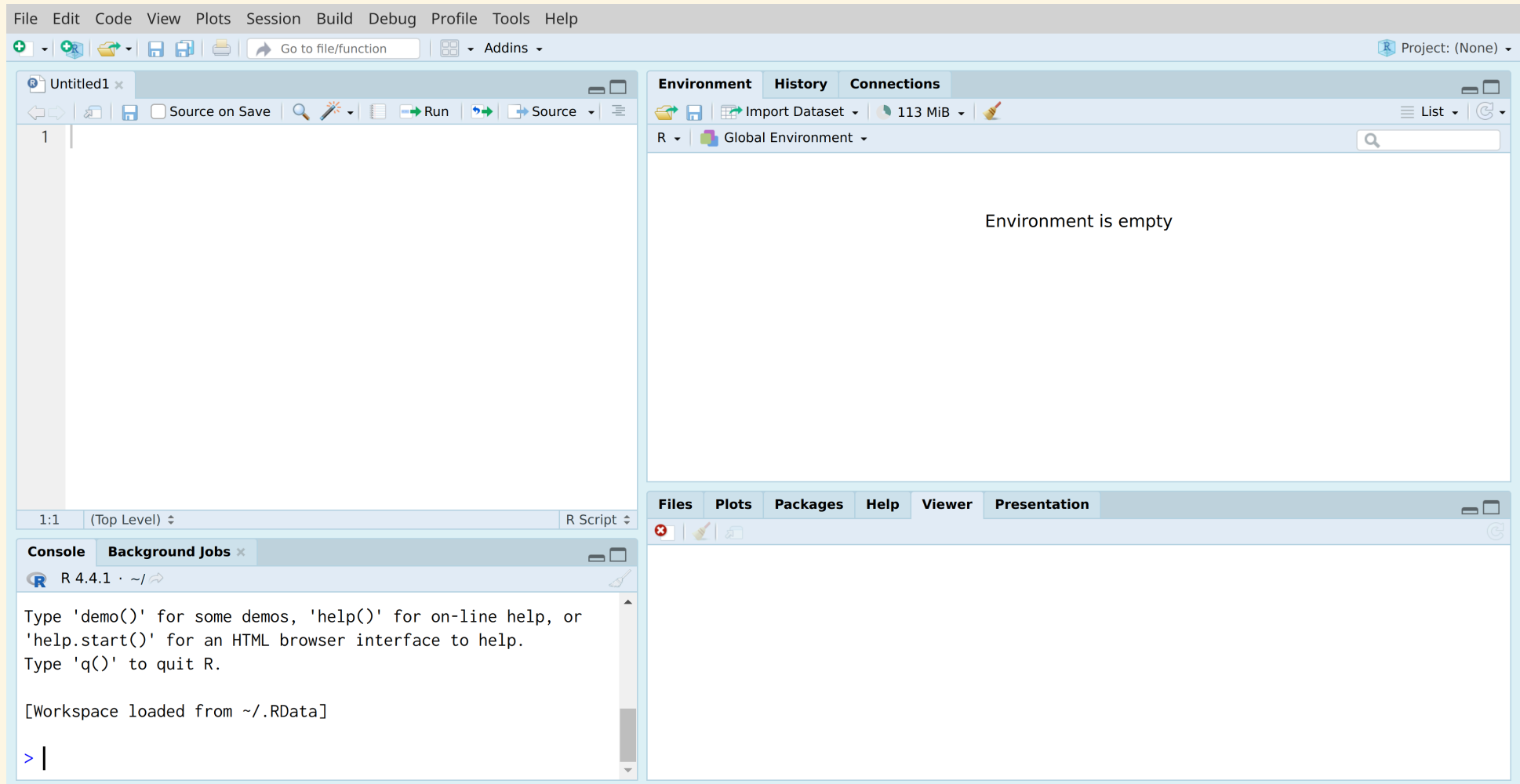
The console is like a message thread

- `code` = messages you write to your computer
- `output` = messages you receive from your computer

R versus RStudio

- roughly speaking, you can think of R as being the console
- RStudio is the other panes and other features

RStudio's panes



RStudio's panes

- you can rearrange the panes if you wish: Tools > Global Options > Pane Layout

Exercise 1

Exercise 1

1. What is a common term for the `>` symbol in the R console?
2. Does the term “output” refer to messages you write to your computer, or to messages your computer writes to you?
3. How many panes are there in RStudio?

Mathematical operators

Math

- what is 32 plus 27?

Math in R

- what is 32 times 27?
- we can find the solution using the R console:

```
> 32*27  
[1] 864
```

- the asterisk (*) indicates multiplication
- it is an example of a mathematical operator

Mathematical operators

- $\wedge$ for exponentiation
- $*$ for multiplication
- $/$ for division
- $+$ for addition
- $-$ for subtraction

Parentheses

- we can combine operators using parentheses:

```
> (15+3)*2  
[1] 36
```

Order of operations

- your computer will use the standard operator of operations, known as PEMDAS in the United States (or BODMAS in the United Kingdom)

Example: an average

- suppose that Bob has his systolic blood pressure measured twice
 - first measurement = 125 mmHg
 - second measurement = 132 mmHg
- what was his average measurement of systolic blood pressure?

```
> (125+132)/2  
[1] 128.5
```

Example: the area of a circle

- a circle has a radius of 2.15 feet
- what is the area of the circle?

```
> pi*2.15^2  
[1] 14.52201
```

- `pi` is a special value that is built in to R

Spaces

```
> (125+132)/2  
[1] 128.5  
> (125 + 132) / 2  
[1] 128.5
```

- it is fine to include spaces
- doing so may help improve readability
- in fact, many people suggest including spaces before and after mathematical operators, as I'll mention in Session 3

Extra parentheses

```
> 15*3+5*18  
[1] 135  
> (15*3)+(5*18)  
[1] 135
```

- it is fine to include extra parentheses
- doing so may help improve readability

Leading zeros

- for numbers between -1 and 1, you do not need to include a zero before the decimal point:

```
> 0.2 + 0.5  
[1] 0.7  
> .2 + .5  
[1] 0.7  
> -0.8 + -0.3  
[1] -1.1  
> -.8 + -.3  
[1] -1.1
```

- but, including it may help improve readability

Exercise 2

Exercise 2

1. Type, `3-` in the console and press the `return` or `enter` key on your keyboard. Press `return` or `enter` again and again and again. Notice that the console is displaying a `+` sign rather than the usual `>` symbol. The `+` sign indicates that your computer thinks you're not done with your instruction. It's occurring here because `3-` suggests an incomplete instruction to do subtraction. To complete the instruction, type in any number (for example, `19`). You should see the prompt again.

Exercise 2

2. Type `3-` in the console again and press the `return` or `enter` key on your keyboard. Press `return` or `enter` again and again and again. Notice the `+` signs. This time, press `esc` on your keyboard to cancel the instruction. You should see the prompt again.

Exercise 2

3. Suppose that there are three siblings, with heights of 5' 3", 5' 8", and 5' 6". Find the average height of the siblings, in inches, by adding the three heights and dividing by 3. Try to do this all in one line, without using any mental math.

Mathematical functions

Example: square root

- what is the square root of 81?

```
> sqrt(81)
[1] 9
```

- `sqrt`: the **name** of an **R function** (more on this in a moment), indicating that we want to take the square root of a number
- **the parentheses**: a space to put the number in
- **81**: the number we want the square root of

What are functions?

```
> sqrt(81)
[1] 9
```

- you can think of functions as **specific requests**
- with the `sqrt()` function, the request is, conceptually: “hi computer, find the square root of this number, or this set of numbers”

What are functions?

- R comes with many built-in functions
- packages also include functions
- it is possible to write your own functions

How do I use functions?

- when using functions in R, you must include both parentheses
- like passwords, functions are case sensitive
 - for example, `elephant` $\neq$ `ELEphaNT`

Some mathematical functions

- `sqrt()` for square roots
- `abs()` for absolute values
- `exp()` for powers of e

Examples

```
> sqrt(144)
[1] 12
> abs(2-12)
[1] 10
> exp(0.87)
[1] 2.386911
```

Exercise 3

Exercise 3

```
> 49*3  
[1] 147  
> sqrt(49)  
[1] 7
```

1. Thinking about pattern recognition, what do you notice as differences between mathematical operators and mathematical functions?

Object assignment

Example: cancer prevalence

- suppose that you work at the Ministry of Health for a specific country
- for the country's annual report, you will calculate the prevalence of several types of cancer among adult women

Example: cancer prevalence

- you will use the following data:
 - your country has 27,113,508 adult women
 - 12,312 people in the country are living with breast cancer
 - 9,120 people in the country are living with cervical cancer
- you will report the prevalence per 100,000 adult women (10^5)

Example: cancer prevalence

```
> 12312 / 27113508 * 10^5  
[1] 45.4091  
> 9120 / 27113508 * 10^5  
[1] 33.63637
```

- this code is correct, but:
 - it is **tedious** to type **27113508** twice
 - typing **27113508** twice introduces the possibility of **human error**
 - the estimated population size **might later change**
- a solution is to **store the number** 💡

Storing the number

```
> popsize<-27113508
```

- 27113508: a number I want to store
- < -: a left arrow, indicating that I want to store the number
- popsize: a name that I picked that I can use to refer to the number

The jar analogy

```
> popsize<-27113508
```

- 27113508: what I want to put in a jar 🍲
- <-: the action of putting something in a jar ↩️ 🫙
- popsize: a label that I place on the jar 🏷️

The jar analogy



The process is like putting something in a jar, and labeling the jar.

Terminology

```
> popsize<-27113508
```

- the “jar” (`popsize` in this example) is known as an **object**
- the process of putting something in a jar is known as **object assignment**

More terminology: in our example, 27113508 is the object's value and `popsize` is the

Object names

- try to use descriptive object names

Object names

- object names are case sensitive
- object names should begin with a letter
- object names can include _ or .

It is actually possible to define an object name that doesn't begin with a letter, but it is

Object retrieval

```
> popsize  
[1] 27113508
```

Math with objects

```
> 12312 / popsize * 10^5  
[1] 45.4091  
> 9120 / popsize * 10^5  
[1] 33.63637
```

- this is, admittedly, still a little tedious, but:
 - it removes the necessity of memorizing the population size
 - it reduces the likelihood of human error

Can I use the equal sign instead?

- yes:

```
> x<-3  
> x  
[1] 3  
> y=5  
> y  
[1] 5
```

- I never use the equal sign for object assignment, for a couple reasons
- but you can if you want to!

Can I replace objects?

- yes, and notice that **your computer will not warn you about it:**

```
> x<-100  
> x  
[1] 100  
> x<-5  
> x  
[1] 5
```

What's so great about object assignment?

- object assignment is very powerful:
 - for example, you can use it to **clean data**
 - for example, you can use it to **store regression models**
- next week, we will use object assignment to **load a data file** into R

Exercise 4

Exercise 4

1. A common upper threshold for “optimal” LDL cholesterol is 100 mg/dL. Store this value using object assignment, giving the object some arbitrary name, like `LDL.cut` or `ldl_optimal_upper`.
2. Retrieve the object at the console, to verify that the object assignment worked.
3. Use the object you defined to calculate how much higher a measurement of 162 mg/dL is than the optimal threshold.